enormous growth in these complex financial instruments that has overwhelmed our financial markets and superseded the capitalization and trading volume of the investments themselves.

At the same time, the markets have been flooded by so-called derivatives, that is, instruments whose values are derived from yet other financial instruments. (Recall our earlier discussion of the futures and options on the S&P 500 Index.) Through interest rate swaps and credit default swaps (don't ask)—traded back and forth around the globe in nanoseconds—these derivatives are used to assume risk, to magnify risk, and (paradoxically) to hedge risk. Their trading volumes are staggering (though rarely disclosed), and their dimension is grotesquely disproportionate to the instruments from which their value is derived. (The credit obligations subject to default swaps are valued at $2 trillion; the swaps themselves total $62 trillion.) The notional principal value of all derivatives is almost beyond imagination—some $600 trillion, nearly 10 times the $66 trillion gross domestic product (GDP) of the entire world.

The innovation of derivatives has enriched the financial sector (and the rating agencies) with their enormous fees, even as the overrated, as it were, CDOs have wreaked havoc on the balance sheets of those who purchased them and have now been left holding the bag, surprisingly including the banks and brokers that created and sold them. Since virtually all of the main mutual fund managers also run pension money, their broad embrace of CDOs has in addition eroded the retirement plans of tens of millions of citizens.

Not to be outdone, SIVs have also created havoc. For it turns out that in order to sell these instruments to their customers, banks increasingly issued so-called liquidity puts to buyers, effectively guaranteeing to repurchase the SIVs on demand at face value. Citigroup, it turns out, was holding not only $55 billion of CDOs on its books, but also some $25 billion of SIV assets that could be (and later were) "put" back to the bank, a risk not publicly disclosed by Citigroup until November 4, 2007.

Astonishingly, Robert Rubin, chairman of Citigroup's executive committee (and a man, one might say, of not inconsiderable financial acumen), has stated that until the summer of 2007, *he had never*